Plug-and-Play Latent Diffusion for Quantitative Ultrasound Imaging

Ruizhi Zhang, Rui Guo, Yi Zhang, Yhonatan Kvich, Yonina C. Eldar

***Abstract*—Ultrasound (US) full waveform inversion enables high-resolution quantitative imaging that can improve diagnosis. Existing US full waveform inversion approaches produce a deterministic point estimate using either explicit regularization in model-based optimization or supervised learned or unrolled networks trained on paired ground truth data, both of which limit flexibility in probabilistic prior modeling and robustness under measurement variations. This paper introduces a posterior sampling approach based on latent diffusion for US inverse, adapted from a generative plug-and-play (PnP) posterior sampling framework. This approach allows to flexibly integrate prior knowledge into physics-based inversion without requiring paired measurement label datasets. Specifically, we first learn the prior distribution of targets from an unlabeled dataset, and then incorporate the learned prior into posterior sampling. In addition, our technique leverages a latent diffusion model on speed of sound (SoS) maps to capture their distribution. Subsequently, we perform posterior sampling by alternating between two samplers that respectively enforce the likelihood and prior distributions, based on given measurement and the forward model describing US wave physics. Experimental results on edema diagnosis demonstrate that our approach achieves state-of-the-art performance in reconstruction accuracy and structural similarity while maintaining high measurement fidelity, particularly under limited measurement conditions.**

***Index Terms*—Ultrasound inverse problems, posterior sampling, latent diffusion, plug-and-play.**

I. Introduction

Quantitative ultrasound inversion has been increasingly developed to reconstruct and visualize tissue-related physical parameters such as speed of sound (SoS) and density, offers significant benefits across various applications, such as edema detection, and rapid stroke imaging [1]–[4]. Diagnostic ultrasound (US) imaging is valued for its low cost, non-invasive, real-time imaging capabilities [5], and the absence of ionizing radiation risks associated with other diagnostic techniques like X-ray and CT imaging. However, conventional US imaging is inherently non-quantitative, with limitations in resolution and contrast, and lacks physical interpretability since it cannot reconstruct essential properties. This challenge becomes more pronounced in limited measurement scenarios.

A widely studied approach is full-waveform inversion (FWI), which estimates quantitative parameters by matching measured and simulated waveforms under a wave propagation model [6]. While FWI can be highly informative, it is also challenging due to strong non-linearity and non-convexity, often requiring many iterations. To improve practicality, learning-assisted methods have been recently proposed [7]. Model-based quantitative radar and US (MB-QRUS) is a representative approach, which incorporates the physical forward model and learns a gradient-related update with a neural network, aiming at real-time reconstruction of multiple physical property maps [8]. Building on this direction, deep unfolded FWI (DUFWI) was further proposed to learn optimized gradient step in multiple iterations, with reported gains in reconstruction quality and robustness, while reducing the number of iterations required by classical FWI [9], [10].

Despite these advances, waveform-based US inversion can remain ill-conditioned under limited measurements [11]. This motivates approaches that incorporate stronger and more expressive priors than handcrafted regularizers, while still enforcing measurement consistency through the governing physics [12]. Plug-and-play (PnP) inference offers a natural mechanism to achieve this by coupling a physics-driven data-consistency operator with a learned prior module [13]–[15]. In particular, a recent latent diffusion PnP (L-DPnP) framework for electromagnetic inverse scattering and brain imaging was proposed in [16], which learns a prior from unlabeled target maps, and then alternates between enforcing the likelihood induced by the forward model and enforcing the learned prior in a compact latent space to improve tractability.

Motivated by [16], we adapt L-DPnP to US inversion by coupling an US forward model-based data-consistency step with a latent diffusion prior learned from properties. Our contributions are as follows: we formulate an L-DPnP approach for US reconstruction, develops an inference procedure that keeps explicit physical consistency while handling the non-linearity of original problem, and demonstrates improved reconstruction fidelity and robustness over existing methods.

This paper is organized as follows: Section II formulates the forward model and the associated inverse problem. Section III presents our LDPnP approach to address the original problem. Implementation details and experimental results are shown in Section IV and Section V, respectively. Section VI offers concluding insights and discusses potential future directions.

II. Problem Formulation

As shown in Fig. 1, we consider a typical 2-D US imaging setup, where a target domain is surrounded by N US transducers. In this configuration, each transducer transmits a waveform sequentially, while some of remaining transducers serve as receivers to record the resulting pressure field. Under

Ruizhi Zhang is with University of Electronic Science and Technology of China, China, and also with Faculty of Math and Computer Science, Weizmann Institute of Science, Israel. Rui Guo, Yi Zhang, Yhonatan Kvich, Yonina C. Eldar are with Faculty of Math and Computer Science, Weizmann Institute of Science, Israel.

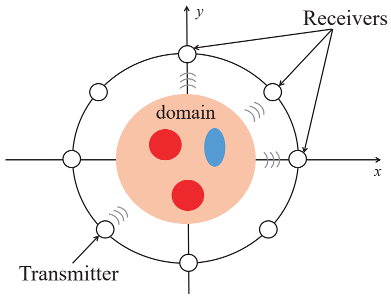

Fig. 1. A typical 2-D measurement setup for US imaging, illustrating a target surrounded by a circular array of transducers.

a given SoS distribution, the wave propagation from the transmitter to the receivers can be modeled using a physics-based forward model, i.e., the acoustic wave equation [17]. By solving this equation via a time-domain finite-difference method with boundary conditions, we can simulate the pressure field and the corresponding received measurements [11].

We discretize the domain into $N_x \times N_y$ subunits, resulting in a discrete SoS map $\mathbf{s} \in \mathbb{R}^{N_x \times N_y}$. The pressure fields at the receivers are sampled after numerical computation. Assuming there are N_T transmissions, with each transmission having N_R transducers receiving the scattered field, and the number of time samples is N_M, the forward process can be given by

$$\mathbf{d} = F(\mathbf{s}) + \mathbf{n}, \tag{1}$$

where $\mathbf{d}$ is the channel data vector of size $N_T N_R N_M$, $F(\cdot)$ is the forward modeling operator that simulates the pressure field given a SoS map, and $\mathbf{n}$ denotes measurement noise.

Our goal is to reconstruct the $\mathbf{s}$ from channel data $\mathbf{d}$. From a probabilistic viewpoint, this problem can be formulated as a Bayesian inverse problem, where $\mathbf{d}$ and $\mathbf{s}$ are treated as random variables, and prior distributions on $\mathbf{s}$ constrain the solution space. In the remainder of the paper, we aim to draw samples from the posterior $p(\mathbf{s}|\mathbf{d})$, which is proportional to

$$p(\mathbf{s}|\mathbf{d}) \propto p(\mathbf{d}|\mathbf{s}) \cdot p(\mathbf{s}), \tag{2}$$

where $p(\mathbf{d}|\mathbf{s})$ denotes the likelihood induced by the forward model (1), and $p(\mathbf{s})$ is the prior distribution capturing statistical structure and physical constraints. A main difficulty is that $p(\mathbf{s})$ is not closed-form. The following section presents how to learn the prior and then perform posterior sampling accordingly.

III. L-DPnP Framework

The L-DPnP framework consists of a pre-training stage and an inversion stage, as shown in Fig. 2. In the pre-training stage, an autoencoder is trained on a SoS dataset $\mathbf{s}$ to obtain a latent representation $\mathbf{z}$. And a latent diffusion model is then trained on the $\mathbf{z}$-space to learn space using an iterative sampling procedure. In each iteration, there is a likelihood step that promotes measurement fitness, and a prior step that encourages agreement with the learned prior.

A. Latent representation

Representing $\mathbf{s}$ with discrete pixels results in high dimensionality and redundant details. To overcome this issue, we project $\mathbf{s}$ onto the latent space of an autoencoder, allowing to represent the distribution of $\mathbf{s}$ in a compact dimensional space.

We train a convolutional autoencoder on a SoS map dataset. The autoencoder contains an encoder and decoder, denoted as $\mathcal{E}$ and $\mathcal{G}$, respectively [18]. The encoder learns to map the input $\mathbf{s}$ to a low-dimensional latent variable $\mathbf{z}$ as

$$\mathbf{z} = \mathcal{E}(\mathbf{s}) \in \mathbb{R}^{N_u \times N_v} \tag{3}$$

with $N_u < N_x$ and $N_v < N_y$. The decoder learns to reconstruct the original SoS map from $\mathbf{z}$, that is,

$$\mathbf{s} = \mathcal{G}(\mathbf{z}). \tag{4}$$

The training of the autoencoder can be achieved by a self-supervised manner to minimize the reconstruction error.

B. Inverse problem in the latent space

Based on (2) and (4), the posterior of $\mathbf{z}$ under given $\mathbf{d}$ is proportional to

$$p(\mathbf{z}|\mathbf{d}) \propto p(\mathbf{d}|\mathbf{z}) \cdot p(\mathbf{z}), \tag{5}$$

where $p(\mathbf{z})$ is induced by $p(\mathbf{s})$ and the encoder $\mathcal{E}$. We assume $p(\mathbf{z})$ is known here; the learning process of $p(\mathbf{z})$ is explained in Section IV-B. The likelihood $p(\mathbf{d}|\mathbf{z})$ is derived from the following forward process

$$\mathbf{d} = F_{\mathcal{G}}(\mathbf{z}) + \mathbf{n}, \tag{6}$$

where $F_{\mathcal{G}}(\cdot)$ denotes a nonlinear map composed with the pre-trained decoder $\mathcal{G}$. Assuming the measurement noise is Gaussian with variance σ^2, the likelihood $p(\mathbf{d}|\mathbf{z})$ is defined as

$$p(\mathbf{d}|\mathbf{z}) \propto \exp(\mathcal{L}(\mathbf{z}; \mathbf{d})) \tag{7}$$

where $\mathcal{L}(\mathbf{z}; \mathbf{d}) = -\frac{1}{2\sigma^2}||\mathbf{d} - F_{\mathcal{G}}(\mathbf{z})||^2$ represents the data-fidelity. We can transform our problem to sampling the latent variable $\mathbf{z}$ from the posterior (5)

C. Posterior sampling

To draw samples from the posterior distribution in (5) using the likelihood (7) and the prior $p(\mathbf{z})$, we can construct a Markov chain (MC) by alternately sampling from two proximal distributions-one derived from the prior and the other from the likelihood-converges to a stationary distribution that approximates the posterior [19]. Specifically, the two proximal distributions of $p(\mathbf{d}|\mathbf{z})$ and $p(\mathbf{z})$ are given by

$$q_0(\mathbf{z}|\mathbf{v}; \eta) \propto \exp\left(\mathcal{L}(\mathbf{z}; \mathbf{d}) - \frac{1}{2\eta^2}||\mathbf{z} - \mathbf{v}||^2\right), \tag{8}$$

$$q_1(\mathbf{z}|\mathbf{v}; \eta) \propto p(\mathbf{z}) \exp\left(-\frac{1}{2\eta^2}||\mathbf{z} - \mathbf{v}||^2\right), \tag{9}$$

where $\mathbf{v}$ is a random vector and η is a parameter of the proximal distribution. When the new state of the MC is generated by sampling (8) and (9) conditioned by the previous state, the distribution of the samples are close to the posterior as the chain is infinitely long and $\eta \to 0$.

Given an iterate $\hat{\mathbf{z}}_k$ at the k-th iteration, a new sample $\hat{\mathbf{z}}_{k+1}$ is generated by sequentially drawing from two distributions

1) **Likelihood step:**

$$\hat{\mathbf{z}}_{k+\frac{1}{2}} \sim q_0(\mathbf{z}|\hat{\mathbf{z}}_k; \eta_k), \tag{10}$$

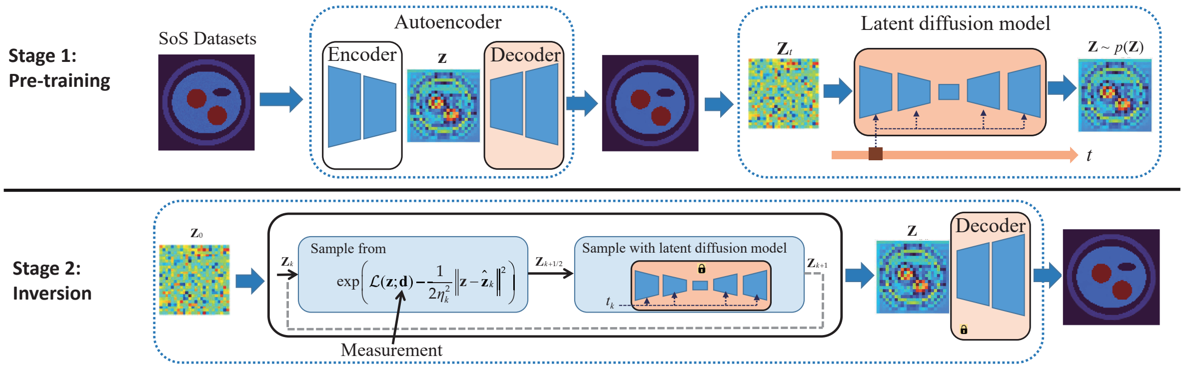

Fig. 2. Overview of the L-DPnP framework.

2) **Prior step:**

$$\hat{\mathbf{z}}_{k+1} \sim q_1(\mathbf{z}|\hat{\mathbf{z}}_{k+\frac{1}{2}};\eta_k), \tag{11}$$

where η_k is an annealing parameter in iterations. The likelihood step and prior step respectively promote consistency with the measurements and the prior knowledge.

In each iteration, given the measurement $\mathbf{d}$ and the iterate $\hat{\mathbf{z}}_k$, the likelihood step (8) generates a sample $\hat{\mathbf{z}}_{k+\frac{1}{2}}$ that promotes measurement fitness $\mathcal{L}(\mathbf{z};\mathbf{d})$. This step can be viewed as sampling from an approximate posterior distribution where the prior $p(\mathbf{s})$ is replaced by a Gaussian distribution concentrated around the iterate $\hat{\mathbf{z}}_k$.

After $\hat{\mathbf{z}}_{k+\frac{1}{2}}$ is obtained, the prior step (9) draws $\hat{\mathbf{z}}_{k+1} \sim q_1(\mathbf{z}|\hat{\mathbf{z}}_{k+\frac{1}{2}};\eta_k)$. This step can be rigorously characterized by the denoising processing using a diffusion model [14]. The second term of (9), $\exp\left(-\frac{1}{2\eta_k^2}||\hat{\mathbf{z}}_{k+\frac{1}{2}}-\mathbf{z}||^2\right)$, can be regarded as a likelihood that is Gaussian

$$p(\hat{\mathbf{z}}_{k+\frac{1}{2}}|\mathbf{z}) \propto \exp\left(-\frac{1}{2\eta_k^2}||\hat{\mathbf{z}}_{k+\frac{1}{2}}-\mathbf{z}||^2\right). \tag{12}$$

Consequently, $q_1(\mathbf{z}|\hat{\mathbf{z}}_{k+\frac{1}{2}};\eta_k)$ is proportional to

$$p(\mathbf{z})p(\hat{\mathbf{z}}_{k+\frac{1}{2}}|\mathbf{z}) \propto p(Z=\mathbf{z}|Z+\eta_k\mathbf{n}=\hat{\mathbf{z}}_{k+\frac{1}{2}};\eta_k^2), \tag{13}$$

with $\mathbf{n} \sim \mathcal{N}(0,\mathbf{I})$ where $\mathcal{N}$ denotes the Gaussian distribution. The expression (13) reformulates (9) as a denoising problem. Specifically, the input $\hat{\mathbf{z}}_{k+\frac{1}{2}}$ is interpreted as a noisy sample corrupted with Gaussian noise of variance η_k^2, and the prior step acts as a denoiser to produce a clean SoS map prediction.

IV. Implementation Details

A. Likelihood step

The likelihood step is achieved using the Langevin dynamics algorithm [20], which transforms the sampling problem into solving a stochastic differential equation (SDE) as follows

$$d\mathbf{z}_\tau = -\nabla\mathcal{L}(\mathbf{z}_\tau;\mathbf{d})d\tau + \frac{1}{\eta_k^2}(\mathbf{z}_\tau - \hat{\mathbf{z}})d\tau + \sqrt{2}d\mathbf{w}_\tau, \tag{14}$$

where τ is the time notation, and $\mathbf{w}_\tau$ represents a standard Wiener process. We apply the exponential integrator to discretize (14) since the linear drift $\frac{1}{\eta_k^2}(\mathbf{z}_\tau - \hat{\mathbf{z}})$ may become very large [15]. In the discrete domain, we represent the time as $\tau_n = n\gamma$, where $n = 0, ..., N_\tau$ with γ being the time interval, and the corresponding discrete variable $\mathbf{z}_\tau$ is denoted as $\mathbf{z}[n]$. Based on (14), the sample at τ_{n+1} is computed by

$$\begin{aligned}\mathbf{z}[n+1] = &\eta_k^2(1-r)\nabla_{\mathbf{z}[n]}\mathcal{L}(\mathbf{z}[n];\mathbf{d}) + r\mathbf{z}[n] \\ &+ (1-r)\hat{\mathbf{z}}_k + \eta_k\sqrt{(1-r)^2}\mathbf{n},\end{aligned} \tag{15}$$

where $r = e^{-\gamma/\eta_k^2}$ and $\mathbf{n} \sim \mathcal{N}(0,\mathbf{I})$. We initialize $\mathbf{z}[0] = \hat{\mathbf{z}}_k$, and after N_τ iterations of the update, we obtain $\mathbf{z}[N_\tau]$. We can obtain this result as $\hat{\mathbf{z}}_{k+\frac{1}{2}} = \mathbf{z}[N_\tau]$.

B. Diffusion models for prior step

Based on III-C, the prior step acts as a denoiser that recovers a clean sample from a noise [14]. Diffusion models learn the prior by training neural networks to invert a progressive noising process, which consists of a perturbation process and corresponding reverse process. Specifically, data samples are gradually corrupted over $t \in [0,1]$ following the SDE

$$d\mathbf{z}_t = f(t)\mathbf{z}_t dt + g(t)d\mathbf{w}_t, \tag{16}$$

where $\mathbf{z}_t$ is the trajectory of random variables in a stochastic process, $f(t)$ and $g(t)$ is the drift and diffusion coefficient of the SDE, respectively, and $\mathbf{w}_t$ is a standard Wiener process. At $t=0$, $\mathbf{z}_0 \equiv \mathbf{z}$, and distribution $p_0(\mathbf{z})$ represents the prior $p(\mathbf{z}) : p_0(\mathbf{z}) \equiv p(\mathbf{z})$. The SDE gradually transforms $\mathbf{z}_0$ into noise, with transition distribution

$$p_{0t}(\mathbf{z}_t|\mathbf{z}_0) = \mathcal{N}\left(\mathbf{z}_t|\alpha(t)\mathbf{z}_0, \beta^2(t)\mathbf{I}\right). \tag{17}$$

Here, $\alpha(t)$ and $\beta(t)$ are determined by $f(t)$ and $g(t)$ [21]. For sampling, the reverse process maps a noisy state $\mathbf{z}_t$ back to a data sample $\mathbf{z}_0 \sim p_0(\mathbf{z})$ by solving the reverse-time SDE

$$d\mathbf{z}_t = \left(f(t)\mathbf{z}_t - g^2(t)\nabla_{\mathbf{z}_t}\log p_t(\mathbf{z}_t)\right)dt + g(t)d\overline{\mathbf{w}}_t, \tag{18}$$

where dt and $\overline{\mathbf{w}}_t$ respectively denotes the time differential and a standard Wiener process running in reverse time. The score function denoted by $\nabla_{\mathbf{z}_t}\log p_t(\mathbf{z}_t)$ is approximated training a time-dependent score model $\mathcal{S}(\mathbf{z}_t, t)$ using denoising score matching on N_{data} samples $\{\mathbf{z}_0^{(i)}\}_{i=1}^{N_{\text{data}}} \sim p(\mathbf{z})$

$$\begin{aligned}\min \frac{1}{N_{\text{data}}}\sum_{i=1}^{N_{\text{data}}} &\mathbb{E}_{t\sim\mathcal{U}[0,1]}\mathbb{E}_{\mathbf{z}_t^{(i)}\sim p_{0t}(\mathbf{z}_t^{(i)}|\mathbf{z}_0^{(i)})} \\ &\left[||\mathcal{S}(\mathbf{z}_t^{(i)},t) - \nabla_{\mathbf{z}_t^{(i)}}\log p_{0t}(\mathbf{z}_t^{(i)}|\mathbf{z}_0^{(i)})||_2^2\right],\end{aligned} \tag{19}$$

where $\mathcal{U}[0,1]$ is the uniform distribution on $[0,1]$, and $p_{0t}(\mathbf{z}_t^{(i)} \mid \mathbf{z}_0^{(i)})$ denotes the transition distribution in (17). In practice, the clean sample $\mathbf{z}_0$ is produced by the encoder $\mathcal{E}$, and the corresponding noisy $\mathbf{z}_t$ is obtained by sampling from $p_{0t}(\mathbf{z}_t \mid \mathbf{z}_0)$. After training, the score network satisfies $\mathcal{S}(\mathbf{z}_t, t) \approx \nabla \mathbf{z}_t \log p_t(\mathbf{z}_t)$. Substituting it into (18) and starting from $\mathbf{z}_T$ at $T \in [0,1]$ yields the reverse SDE for sampling.

V. Simulation Results

A. Datasets and Setup

To evaluate the L-DPnP, we generated a simulated arm dataset that mimics the cross-sectional anatomy of a human arm. Each sample is defined on a $10\ \mathrm{cm} \times 10\ \mathrm{cm}$ domain and contains an elliptical foreground region with assigned tissue properties. The outer layer represents skin, followed by subcutaneous fat and an inner muscle compartment. Two circular bones are embedded within the muscle, and a small elliptical edema region is optionally included. SoS values were chosen according to physiological measurements: muscle (1570–1620 m/s), fat (1420–1450 m/s), skin (1530–1560 m/s), bone (2700–3000 m/s), and edema (1450–1500 m/s) [22]. We used a unified dataset containing both edema-present and edema-absent samples. In total, 30,000 SoS maps were generated, with 24,000 for training and 6,000 for validation.

The circular array comprises 16 uniformly spaced transducers. Each transducer transmits once in a round manner (16 transmissions in total). For each transmission, one element acts as the transmitter, while three elements serve as receivers: the diametrically opposite element and its two neighbors, yielding 3 channels per transmission and thus 48 channels overall. Each element emits a Gaussian pulse with a center frequency of 235 kHz. The region of interest spans 94×94 pixels.

We compared our method with traditional FWI, as well as the recent model-driven DL approach MB-QRUS and DUFWI. FWI use the same channel data as our method, optimized via L-BFGS with the data-misfit loss. For MB-QRUS and DUFWI, we re-trained the U-Net in [8] and the DUFWI networks in [9], respectively, using data generated from SoS maps dataset and forward model. FWI requires one forward computation per iteration and typically hundreds of iterations. MB-QRUS performs a single update (one forward computation). DUFWI significantly reduces forward computations compared to FWI at higher training cost. L-DPnP usually needs more forward computation, but it can learn property-domain priors and reduce training overhead by avoiding repeated full-wave simulations across configurations. The numbers of iterations in FWI and DUFWI are set to 100 and 5, respectively. For L-DPnP, we run 30 iterations, where each iteration consists of 500 likelihood steps and 300 denoising steps.

B. Results

Fig. 3 compares the reconstructed SoS maps of the simulated arm data using FWI, MB-QRUS, DUFWI, and the proposed L-DPnP, with the ground truth shown in the last column. FWI suffers from severe speckle artifacts and fails to reliably recover the bone region and the low-speed skin boundary or the edema region in the edema case. MB-QRUS effectively reduces speckle noise but exhibits noticeable over-smoothing, which blurs tissue interfaces and weakens the visibility of the edema. DUFWI improves the recovery of the edema region and provides sharper structures than FWI and MB-QRUS and higher reconstruction fidelity, although residual bias and structural variations can still be observed. Specifically, the proposed L-DPnP achieves the most accurate reconstructions across all cases, producing sharper tissue boundaries and better matching the ground truth while maintaining high background fidelity. This demonstrates that the PnP strategy can robustly incorporate strong image priors into the inversion process, leading to improved structural consistency and enhanced reconstruction reliability. However, the skin region is still not perfectly recovered, indicating remaining ambiguity under the limited measurements. This can be attributed to the fact that sampling-based methods can explore multiple plausible inversions that yield similar data misfit.

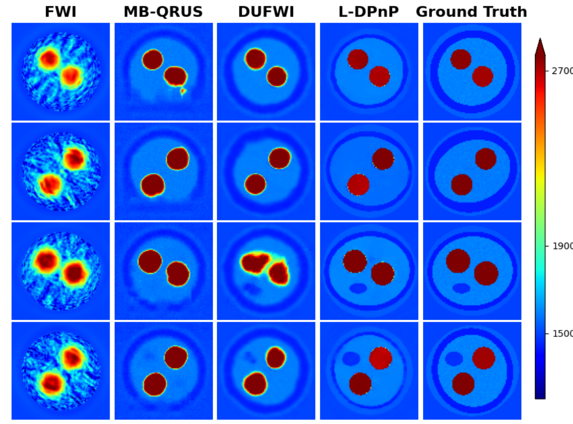

Fig. 3. Reconstructed SoS maps from the simulated arm data.

TABLE I
QUANTITATIVE COMPARISON FOR THE RECONSTRUCTED SOS MAPS OF SIMULATED ARM DATA IN FIG. 3.

Metric	FWI	MB-QRUS	DUFWI	L-DPnP
SSIM (↑)	0.5796	0.7762	0.7855	**0.8133**
PSNR (dB) (↑)	20.16	24.30	24.73	**25.18**
NMSE (dB) (↓)	-20.65	-24.66	-22.79	**-24.24**
RMSE (↓)	156.2035	**105.9036**	123.5394	108.6761

The quantitative evaluation is summarized in Table I, where L-DPnP achieves the best overall performance in terms of SSIM, PSNR, and NMSE, confirming its superior structural consistency and global reconstruction fidelity. Although MB-QRUS attains the lowest RMSE, its overly smoothed distribution may reduce structural details, explaining why its SSIM and PSNR remain inferior to L-DPnP. Besides, RMSE is more sensitive to localized outlier errors, which can slightly penalize L-DPnP despite its better overall reconstruction quality.

VI. Conclusion

We present a latent-diffusion posterior sampling framework for ill-posed US inverse problems, combining a data-fidelity likelihood step with a latent prior step. This design allows flexible integration of physical models into generative sampling, yielding reconstructions with low error and high constrast. Experiments show state-of-the-art performance over existing physics-driven and learning-based methods.

References

[1] K. Wang, T. Matthews, F. Anis, C. Li, N. Duric, and M. A. Anastasio, "Waveform inversion with source encoding for breast sound speed reconstruction in ultrasound computed tomography," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 62, no. 3, pp. 475–493, Mar. 2015.

[2] C. Pirri, C. Ferraretto, N. Pirri, L. Bonaldo, R. De Caro, S. Masiero, and C. Stecco, "Ultrasound examination of skin, fasciae and subcutaneous tissue: Optimizing rehabilitation for secondary upper limb lymphedema," *Diagnostics*, vol. 14, no. 24, Art. no. 2824, 2024.

[3] D. H. Lee, "Imaging evaluation of non-alcoholic fatty liver disease: Focused on quantification," *Clinical and Molecular Hepatology*, vol. 23, no. 4, pp. 290–301, 2017.

[4] L. Guasch, O. Calderón Agudo, M.-X. Tang, P. Nachev, and M. Warner, "Full-waveform inversion imaging of the human brain," *npj Digital Medicine*, vol. 3, Art. no. 28, 2020.

[5] T. L. Szabo, *Diagnostic Ultrasound Imaging: Inside Out*, 2nd ed. Academic Press, 2013.

[6] J. Virieux and S. Operto, "An overview of full-waveform inversion in exploration geophysics," *Geophysics*, vol. 74, no. 6, pp. WCC1–WCC26, 2009.

[7] M. Ambrosanio, S. Franceschini, V. Pascazio, and F. Baselice, "An end-to-end deep learning approach for quantitative microwave breast imaging in real-time applications," *Bioengineering*, vol. 9, no. 11, Art. no. 651, 2022.

[8] T. Sharon and Y. C. Eldar, "Real-time model-based quantitative ultrasound and radar," *IEEE Transactions on Computational Imaging*, vol. 10, pp. 1175–1190, 2024.

[9] N. Cohen, Y. Kvich, R. Guo, and Y. C. Eldar, "Deep unfolding of full waveform inversion for quantitative ultrasound imaging," in *Proc. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Hyderabad, India, 2025, pp. 1–5.

[10] R. Guo, T. Huang, M. Li, H. Zhang, and Y. C. Eldar, "Physics-embedded machine learning for electromagnetic data imaging: Examining three types of data-driven imaging methods," *IEEE Signal Processing Magazine*, vol. 40, no. 2, pp. 18–31, Mar. 2023.

[11] A. Shultzman and Y. C. Eldar, "Nonlinear waveform inversion for quantitative ultrasound," *IEEE Transactions on Computational Imaging*, vol. 8, pp. 893–904, 2022.

[12] S. V. Venkatakrishnan, C. A. Bouman, and B. Wohlberg, "Plug-and-play priors for model based reconstruction," in *Proc. IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, 2013, pp. 945–948.

[13] A. Bora, A. Jalal, E. Price, and A. G. Dimakis, "Compressed sensing using generative models," in *Proc. 34th International Conference on Machine Learning (ICML)*, 2017, pp. 537–546.

[14] X. Xu and Y. Chi, "Provably robust score-based diffusion posterior sampling for plug-and-play image reconstruction," in *Advances in Neural Information Processing Systems*, vol. 37, pp. 36148–36184, 2024.

[15] Z. Wu, Y. Sun, Y. Chen, B. Zhang, Y. Yue, and K. L. Bouman, "Principled probabilistic imaging using diffusion models as plug-and-play priors," in *Advances in Neural Information Processing Systems*, vol. 37, pp. 118389–118427, 2024.

[16] R. Guo, Y. Zhang, Y. Kvich, T. Huang, M. Li, and Y. C. Eldar, "Plug-and-play latent diffusion for electromagnetic inverse scattering with application to brain imaging," *arXiv preprint arXiv:2509.04860*, 2025.

[17] Y. Yuan, Y. Zhao, N. Zhang, Y. Xiao, J. Jin, N. Feng, and Y. Shen, "Full-waveform inversion for breast ultrasound tomography using line-shape modeled elements," *Ultrasound in Medicine & Biology*, vol. 49, no. 5, pp. 1070–1081, 2023.

[18] R. Guo, Z. Lin, J. Xin, M. Li, F. Yang, S. Xu, and A. Abubakar, "Three-dimensional microwave data inversion in feature space for stroke imaging," *IEEE Transactions on Medical Imaging*, vol. 43, no. 4, pp. 1365–1376, Apr. 2024.

[19] C. A. Bouman and G. T. Buzzard, "Generative plug and play: Posterior sampling for inverse problems," in *Proc. 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, 2023, pp. 1–7.

[20] G. O. Roberts and J. S. Rosenthal, "Optimal scaling of discrete approximations to Langevin diffusions," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 60, no. 1, pp. 255–268, 1998.

[21] S. Särkkä and A. Solin, *Applied Stochastic Differential Equations*. Cambridge, U.K.: Cambridge University Press, 2019.

[22] F. A. Duck, "Acoustic properties of tissue at ultrasonic frequencies," in *Physical Properties of Tissues: A Comprehensive Reference Book*. London, U.K.: Academic Press, 1990, ch. 4, pp. 73–135.